



Pak-Austria Fachhochschule:

Institute of Applied Sciences & Technology

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Minutes of the GEC meeting for MS Research Work Evaluation of Ms. Sabahat Siddique

The GEC meeting to evaluate MS Research thesis of **Ms. Sabahat Siddique** Reg# M24F0023AI003 was held on **8th January 2026** at 2:00 PM. Ms. Sabahat presented her synopsis in front of the GEC committee.

The following changes have been suggested:

- 1. The base/reference paper is from 2021. Please review and confirm whether any recent work has been carried out in the same research area since then.**

Aspect	Ferrario et al. (2021)	Guizzardi et al. (2021)	Spahn (2023)	Lu et al. (2022)	E-LENS (2025)	Proposed Research (Your Work)
Focus Area	Ethical requirements identification	RE-based ethical modeling	Ontology-based ethics	Responsible AI roadmap	Ethics assurance	Ethical RE for AI (end-to-end)
Ethical Principles Covered	Fairness, accountability, transparency	Ethics in RE	Ethicality concepts	High-level ethics	Ethics validation	Fairness, Privacy, Explainability, Responsibility
RE Lifecycle Coverage	Partial	Partial	Partial	Conceptual only	Partial	Full RE lifecycle (elicitation → traceability)
Measurable Ethical Metrics	No	No	No	No	Limited	Yes (fairness, explainability, privacy metrics)

Dataset-Level Evaluation	No	No	No	No	Limited	Yes (Adult, COMPAS, German Credit, Bank Marketing)
ML Model Integration	No	No	No	No	Limited	Yes (LogReg, RF, XGBoost, etc.)
Explainability Tools (SHAP/LIME)	No	No	No	No	Partial	Fully integrated
Fairness Toolkits (AIF360/Fairlearn)	No	No	No	No	Partial	Fully integrated
Validation Automation	No	No	No	No	Semi-manual	Automated via web tool
Traceability Support	No	No	Ontology only	No	Argument-based	Full traceability (Req → Data → Model → Metric)
Web-Based Tool Support	No	No	No	No	Prototype	Complete validation platform
Industry Readiness	Low	Low	Medium	Low	Medium	High (practical & deployable)
Continuous Monitoring	No	No	No	No	Limited	Yes

2. The matrix of the four ethical requirements must be clearly justified.

- Explain whether and how ethical issues may arise.
- Provide clear examples (e.g., Justice vs. Equity).
- Clearly justify the rules and assumptions used for each ethical requirement.
- Explain the matrix with simple and relevant examples.

COMPLETE ETHICAL REQUIREMENTS MATRIX

Ethical Requirement	Potential Issues & How They Arise	Justice vs. Equity Examples	Rules & Assumptions	Simple Example
FAIRNESS	How Issues Arise: 1. Historical Bias: Training data reflects past discrimination 2. Proxy Discrimination: Using ZIP code as income proxy 3. Algorithmic Amplification: ML amplifies small biases 4. Fairness-Utility Tradeoff: Fairness vs. accuracy conflict	Justice (Equal Treatment): : - Everyone gets same loan approval criteria - Same hiring test for all candidates Equity (Equal Outcome): - Adjust loan criteria for marginalized groups - Different hiring thresholds to achieve diversity	Rules: 1. No discrimination on protected attributes 2. Equal opportunity for qualified individuals 3. Regular fairness audits required Assumptions: - Protected attributes identifiable - Historical disadvantage exists - Merit can be measured objectively	Loan Approval AI: Justice: Same credit score threshold for all; discriminates against historically underbanked groups Equity: Adjust thresholds by neighborhood poverty level : achieves fair outcomes
PRIVACY	How Issues Arise: 1. Data Re-identification: Anonymous data linked back to individuals 2. Function Creep: Data used beyond original purpose	Justice: - Same privacy rules for everyone - Uniform data collection limits	Rules: 1. Data minimization principle 2. Purpose limitation 3. k-anonymity ≥ 5	Health App: Justice: Collect same data from all users : exposes vulnerable populations Equity: Allow vulnerable

	3. Inference Attacks: AI infers sensitive information 4. Consent Fatigue: Users agree without understanding	Equity: • Extra privacy for vulnerable groups • Special protections for marginalized communities	4. User consent required Assumptions: • Users value privacy • Data can be sufficiently anonymized • Purpose can be clearly defined	users to opt-out of sensitive data collection: protects those at risk
TRANSPARENCY (EXPLAINABILITY)	How Issues Arise: 1. Black Box Problem: Complex models uninterpretable 2. Stakeholder Mismatch: Wrong explanation for wrong audience 3. Selective Transparency: Only sharing favorable information 4. Explanation Manipulation: Gameable explanations	Justice: • Same explanation format for all users • Standard technical explanations Equity: • Tailored explanations based on user knowledge • Multiple explanation formats for accessibility	Rules: 1. Must provide meaningful explanations 2. Explanation must match model reasoning 3. Different explanations for different stakeholders Assumptions: • Explanations improve	Medical Diagnosis AI: Justice: Give same technical explanation to all: confuses patients Equity: Give doctors technical details, patients simple visual explanations : appropriate understandings

			trust • Users can understand appropriate explanations • Fidelity can be measured	
ACCOUNTABILITY	How Issues Arise: 1. Responsibility Diffusion: "The algorithm decided" excuse 2. Audit Trail Gaps: Missing decision logs 3. Redress Barriers: Difficult to challenge decisions 4. Legal Loopholes: No liability for AI errors	Justice: • Same appeal process for all • Uniform liability rules Equity: • Simplified redress for disadvantaged users • Proactive monitoring for vulnerable groups	Rules: 1. Clear responsibility assignment 2. Complete audit trails 3. Accessible appeals mechanism 4. Regular impact assessments Assumptions: • Responsibility can be clearly assigned • Logs don't compromise privacy • Redress mechanisms are effective	Hiring AI Rejection: Justice: Standard appeal form: disadvantages non-technical applicants Equity: Multiple appeal channels (phone, online, in-person): ensures access for all

JUSTIFICATION & EXAMPLES

1. FAIRNESS: Justice vs. Equity in Practice

Scenario: University Admissions AI

Approach	Justice (Equality)	Equity (Equality of Outcome)
Method	Same SAT score cutoff for all applicants	Holistic review considering socioeconomic background
AI Implementation	Automatically admit above 1400 SAT	Adjust thresholds based on school quality, neighborhood
Result	Mostly wealthy suburban students admitted	Diverse student body including underrepresented groups
Ethical Issue	Perpetuates existing educational inequalities	May be seen as "reverse discrimination"
Justification	"Merit-based" but ignores unequal starting points	Acknowledges systemic barriers and seeks to correct them

Rules for Fairness:

1. **Group Fairness:** Protected groups should have similar outcomes
2. **Individual Fairness:** Similar individuals should have similar outcomes
3. **Counterfactual Fairness:** Decision shouldn't change if protected attribute changed

Assumption Challenge: Merit is objectively measurable. Reality: "Merit" often reflects privilege.

2. PRIVACY: The Re-identification Problem

Example: "Anonymous" Location Data

Issue	How It Happens	Consequence
Data Collection	App tracks location every 5 minutes	Creates movement pattern
"Anonymization"	Remove name, keep ZIP code and timestamps	Still identifies individuals
Re-identification	Cross-reference with public social media check-ins	Identifies specific person
Harm	Stalker uses data to track victim	Physical safety risk

The Privacy Paradox:

- **Justice:** Same privacy settings for all → doesn't protect vulnerable populations
- **Equity:** Extra privacy protections for domestic violence victims → recognizes differential risks

Rules Justification:

- **k-anonymity ≥ 5 :** Each person indistinguishable from 4 others
- **l-diversity ≥ 2 :** Each group has multiple sensitive attribute values
- **t-closeness:** Sensitive attribute distribution similar to population

3. TRANSPARENCY: The Explanation Spectrum

Medical AI Diagnosis Example:

Stakeholder	Need	Justice Issue	Equity Solution
Patient	"Why do I have cancer?"	Too technical: "95% probability based on 1000 features"	Visual: "These 3 spots on your scan are concerning"
Doctor	"Why this treatment recommendation?"	Too vague: "Model suggests Chemo A"	Detailed: "Based on 5 similar cases with patient's biomarkers"
Regulator	"Is it safe and unbiased?"	No evidence of testing	Complete validation report with demographic breakdown

Explanation Types:

1. **Global:** How model works overall
2. **Local:** Why this specific decision
3. **Counterfactual:** What would change the decision
4. **Contrastive:** Why this vs. alternative

Rule: Explanation must match audience's mental model

Assumption: Users want explanations. Reality: Some prefer simple trust.

4. ACCOUNTABILITY: The Responsibility Chain

Autonomous Vehicle Accident:

Responsibility Level	Traditional (Justice)	AI System (Equity)
Immediate	Driver at fault	System failed to detect pedestrian
Technical	Brake failure	Sensor algorithm error
Organizational	Manufacturer defect	Inadequate testing for diverse conditions
Societal	Individual responsibility	System design prioritizing certain lives

Accountability Mechanisms:

1. **Audit Trails:** Complete decision logs
2. **Redress:** Easy appeal process
3. **Compensation:** Clear liability rules
4. **Prevention:** Regular safety updates

Justice Issue: Same liability for all → ignores power imbalance

Equity Solution: Stricter accountability for corporations vs. individuals

CONFLICT RESOLUTION MATRIX

Conflict	Example	Resolution Principle	Implementation
Fairness vs. Privacy	Can't check for bias without protected attributes	Privacy-preserving fairness	Use differential privacy, federated learning
Transparency vs. Privacy	Explaining decisions reveals sensitive patterns	Selective transparency	Aggregate explanations, hide individual details
Fairness vs. Transparency	Simple models (fair but less accurate) vs. complex (accurate but unfair)	Explainable fairness	Use interpretable fairness-aware models
Accountability vs. Innovation	Strict liability may discourage AI development	Risk-proportionate accountability	Higher stakes = stricter accountability

How our Platform Addresses Each Requirement:

1. Fairness Validation:

- **Input:** Protected attributes, fairness metric choice
- **Test:** Multiple fairness metrics (demographic parity, equal opportunity)
- **Output:** Quantified fairness scores with thresholds
- **Justice/Equity Choice:** User selects which fairness definition to use

2. Privacy Validation:

- **Input:** Dataset, quasi-identifiers, sensitive attributes
- **Test:** k-anonymity, l-diversity, t-closeness

- **Output:** Privacy compliance report with recommendations
- **Justice/Equity:** Can apply stricter privacy to vulnerable groups

3. Transparency Validation:

- **Input:** Model, sample data, stakeholder type
- **Test:** Explanation fidelity, completeness, understandability
- **Output:** Explanation quality scores
- **Justice/Equity:** Different validation for different stakeholder explanations

4. Accountability Validation:

- **Input:** System logs, decision records, audit trails
- **Test:** Completeness, accessibility, redress mechanisms
- **Output:** Accountability score with improvement suggestions
- **Justice/Equity:** Checks for accessibility of appeal mechanisms

Example: AI Grading System

FAIRNESS ISSUE:

- Justice: Same grading rubric for all students
- Problem: Disadvantages ESL students
- Equity: Adjust rubric for language difficulty

PRIVACY ISSUE:

- Justice: Store all student work
- Problem: Exposes personal information
- Equity: Anonymize before storage

TRANSPARENCY:

- Justice: "Grade: B"
- Problem: No learning feedback
- Equity: "Strong argument but needs more evidence"

ACCOUNTABILITY:

- Justice: "Computer says no"
- Problem: No appeal process
- Equity: Human review available

1. Justice vs. Equity is Contextual:

- **Justice:** Treating everyone the same
- **Equity:** Giving people what they need to succeed

- **Our Tool:** Allows selection based on context

2. Interdependence of Requirements:

- No privacy without fairness (biased data collection)
- No accountability without transparency
- No fairness without privacy protection

3. Trade-offs are Inevitable:

- More fairness: Less accuracy (sometimes)
- More privacy: Less personalization
- More transparency: More vulnerability
- More accountability: Higher costs

My Research Contribution:

Before: Ethical requirements as abstract principles

After: Quantified, testable requirements with clear justice/equity distinctions

3- Clearly explain the equations, symbols, and terminology used in the ethical requirements matrix to ensure proper understanding.

Ethical Requirements: Equations & Symbols Explained

1. FAIRNESS METRICS & FORMULAS

Key Symbols:

- TP = True Positives (correctly accepted)
- TN = True Negatives (correctly rejected)
- FP = False Positives (wrongly accepted)
- FN = False Negatives (wrongly rejected)
- A, B = Different groups (e.g., men/women)
- N = Total number
- P = Probability

Common Fairness Metrics:

a) Demographic Parity (Statistical Parity)

Formula: $P(\hat{Y}=1|A) = P(\hat{Y}=1|B)$

What it means: Selection rates should be equal across groups

Example:

- Group A (men): 70% get loans
 - Group B (women): 70% get loans
- Equal = Fair
→ Different = Biased

b) Equal Opportunity

Formula: $P(\hat{Y}=1|Y=1, A) = P(\hat{Y}=1|Y=1, B)$

What it means: True positive rates should be equal

Example:

- Qualified men: 90% get hired
 - Qualified women: 90% get hired
- Equal opportunity

c) Equalized Odds

Formula:

$$P(\hat{Y}=1|Y=y, A) = P(\hat{Y}=1|Y=y, B) \text{ for } y \in \{0,1\}$$

What it means: Both true positive AND true negative rates equal

Example:

- Men: 90% correct hires, 10% wrong rejects
 - Women: 90% correct hires, 10% wrong rejects
- Equalized odds

d) Disparate Impact Ratio

$$\text{Formula: } \min(P(\hat{Y}=1|A), P(\hat{Y}=1|B)) / \max(P(\hat{Y}=1|A), P(\hat{Y}=1|B))$$

What it means: Ratio of selection rates (should be close to 1)

Example:

- Men: 80% selected
 - Women: 60% selected
 - Ratio: $60/80 = 0.75$
- Below 0.8 suggests bias

2. PRIVACY METRICS & FORMULAS

Key Symbols:

- k = Minimum group size for anonymity
- l = Minimum distinct values in group
- QI = Quasi-identifiers (can identify people when combined)
- SA = Sensitive attribute (what we want to protect)

Privacy Guarantees:

a) k-Anonymity

Condition: Each combination of QI appears in $\geq k$ records

Example:

If $k=5$, then every group must have at least 5 people

Table of QI combinations:

Age Range	ZIP Code	Count	
30-35	12345	1	← Violation (only 1 person)
30-35	12345	7	✓ OK ($7 \geq 5$)

b) l-Diversity

Condition: Each group has $\geq l$ distinct values of SA

Example:

If $l=2$, each group must have at least 2 different diseases

Table:

Age Range	ZIP Code	Diseases in Group	
30-35	12345	{Diabetes}	← Violation (only 1 disease)
30-35	12345	{Diabetes, Cancer}	✓ OK (2 diseases)

3. TRANSPARENCY METRICS & FORMULAS

Key Symbols:

- F = Feature importance score
- C = Confidence/Probability score
- R = Relevance score
- S = Similarity score
- E = Explanation

Explainability Metrics:

a) SHAP (SHapley Additive exPlanations) Values

Formula: $\phi_i = \sum_{(S \subseteq N \setminus \{i\})} \left[\frac{|S|!(|N|-|S|-1)!}{|N|!} \times [f(S \cup \{i\}) - f(S)] \right]$

What it means: How much each feature contributes to prediction

Example:

Loan decision:

- Income: +0.3 (increases chance)
- Debt: -0.4 (decreases chance)
- Age: +0.1 (slightly increases)
- Total: -0.0 (rejected)

b) LIME (Local Interpretable Model-agnostic Explanations)

Formula: $\text{argmin}_{(g \in G)} [L(f, g, \pi_x) + \Omega(g)]$

What it means: Creates simple local explanation around specific prediction

Example:

Why was THIS loan rejected?

- Debt-to-income ratio: 60% (too high) ✓
- Credit score: 650 (OK) ✓
- Employment: <1 year (too short) ✓

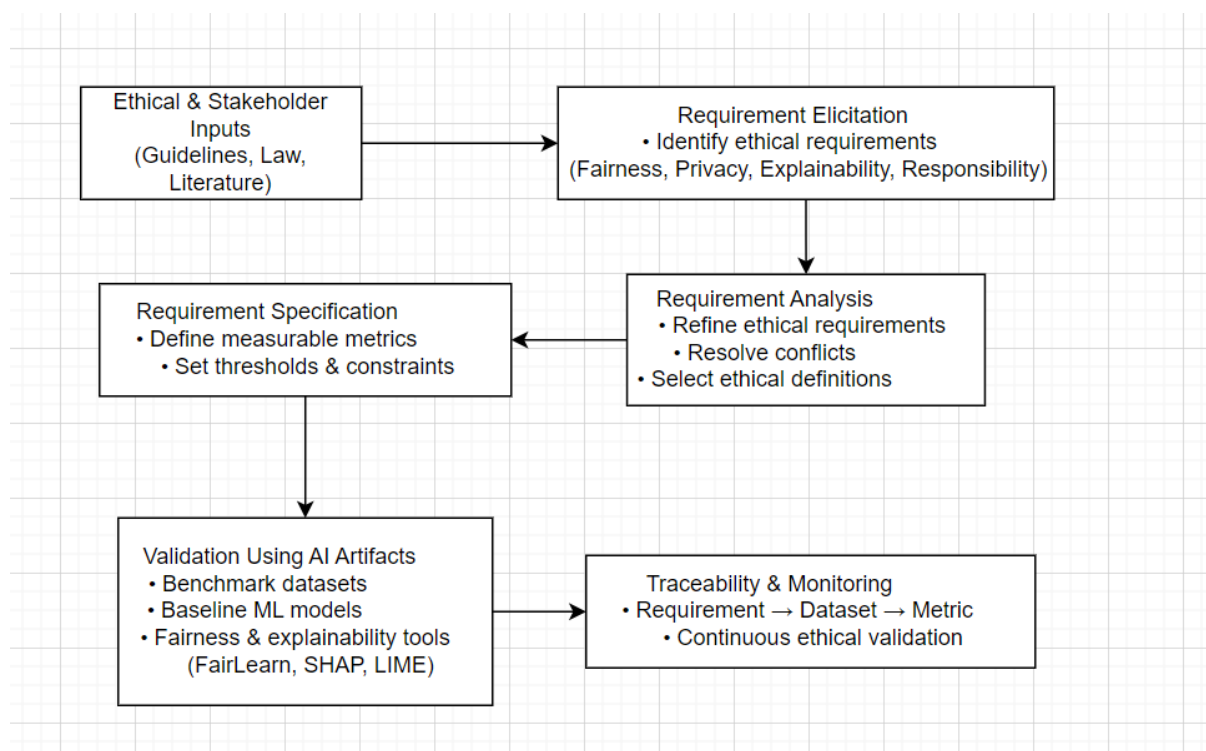
4. The term “RE” in the title should be updated to a term that reflects a phase after system analysis, rather than Requirement Engineering.

Requirement Engineering (RE) is not the implementation itself. In our research, RE is used to define, structure, and validate ethical requirements before and during AI system

development, while datasets and models are used only to validate those requirements, not to replace RE.

This research treats Requirement Engineering as a design-time activity, where ethical requirements are elicited, analysed, and specified prior to AI system development. Machine learning models and benchmark datasets are employed solely to validate these requirements and provide empirical evidence of compliance, without violating the classical RE lifecycle.

- RE defines ethical constraints first.
- Datasets are used to validate requirements.
- Models act as test subjects.



5. Clearly define the scope of the proposed system, including:

- The types of datasets being used
- The models that will be evaluated or verified

Types of Datasets Being Used

This research uses publicly available benchmark datasets that are commonly employed in studies on fairness, bias, and ethical evaluation of AI systems. These datasets contain sensitive attributes (such as gender, age, or race) and decision labels, making them suitable for validating ethical requirements defined during the Requirement Engineering (RE) process. The datasets are used only for validation purposes, not for requirement elicitation.

The types of datasets used include:

1. **Socio-economic Decision Datasets**

These datasets represent real-world decision-making scenarios such as income prediction, credit approval, and marketing responses. They are used to evaluate fairness, accountability, and transparency in AI decision systems.

2. **Bias-Sensitive Classification Datasets**

These datasets explicitly include protected attributes and are widely used to study algorithmic bias and discrimination. They enable the measurement of fairness metrics and validation of fairness-related requirements.

3. **Benchmark Ethical AI Datasets**

Standard datasets adopted by the research community for testing ethical AI frameworks. Using benchmark datasets ensures reproducibility and comparability with existing research.

4. **Structured Tabular Datasets**

All datasets are structured (tabular) and suitable for applying classical machine learning models and fairness evaluation tools.

Why These Dataset Types Are Appropriate

- They support quantitative measurement of ethical requirements
- They include protected attributes needed for fairness analysis
- They are public, well-documented, and reproducible
- They align with the goal of validating ethical requirements defined in RE

Models That Will Be Evaluated or Verified

In this research, baseline and interpretable machine learning models will be evaluated to verify whether the predefined ethical requirements (fairness, privacy, explainability, and responsibility) are satisfied. The models are not proposed as novel AI algorithms but are used as validation instruments within the Requirement Engineering (RE) framework.

The models selected for evaluation include:

1. **Logistic Regression**

Used as a transparent and interpretable baseline model. It enables easy analysis of feature influence and supports explainability and fairness evaluation.

2. **Decision Tree Classifier**

Provides rule-based decision logic that is naturally interpretable. It is useful for validating explainability and accountability requirements.

3. **Random Forest (Baseline Ensemble Model)**

Used to study trade-offs between accuracy and ethical requirements such as fairness and transparency.

4. **Support Vector Machine (Linear Kernel)**

Applied to assess ethical compliance in more complex decision boundaries while maintaining partial interpretability.

Purpose of Model Evaluation

- To verify ethical constraints defined during the RE phase
- To compare ethical trade-offs across different model types
- To provide quantitative evidence for requirement validation
- To demonstrate how ethical requirements can be monitored across models

Models Suitability

- Widely used in ethical AI and fairness research
- Compatible with fairness and explainability tools (FairLearn, SHAP, LIME)
- Computationally efficient and reproducible
- Appropriate for research focused on validation, not model innovation

Important Clarification

The selected models serve as test subjects to validate ethical requirements and do not represent the core contribution of this research.

6- The problem statement should be revised to remove Requirement Engineering, as it does not appropriately fit in this section.

The Word RE is the main alignment of my Research paper. As I already addressed in question 4.

7- All-important terms used in the problem statement should be justified and supported by relevant example.

1. Requirement Engineering (RE) methods don't properly capture ethical needs

Easy meaning: We write down what the AI should DO but forget to write down what the AI should BE.

Like building a car:

- We write: "Car must go 100 mph" (what it should DO)
- We forget: "Car must not crash into people" (what it should BE)

For AI:

- We write: "AI must approve loans quickly"
- We forget: "AI must be fair to all races and genders"

2. Ethical needs: accountability, privacy, explainability, fairness

Accountability = Who's responsible when AI messes up?

- **Easy example:** If a self-driving car crashes, who's responsible?
 - The car company? The programmer? The owner?
 - Currently: Everyone says "not me!"

Privacy = Keeping secrets safe

- **Easy example:** A health app should keep your medical info private
- **Currently:** Apps often share data without clear permission

Explainability = AI should explain its decisions

- **Easy example:** If a loan is denied, the AI should say WHY
- **Currently:** AI often says "no" without explanation

Fairness = Treating everyone equally

- **Easy example:** A hiring AI should not prefer men over women
- **Currently:** Many AI systems show bias

3. "Ethical standards are theoretical and lack metrics"

Theoretical = Just ideas, not rules

- **Easy example:** We all agree "be fair" but nobody says HOW fair
- Like saying "be a good driver" but not defining what that means

Lack metrics = "No numbers to measure"

- **Easy example:**
 - Bad: Be fair (How do we measure this?)
 - Good: Hire equal percentages of men and women ($\pm 5\%$)
 - Currently: We have the "bad" version, not the "good" version

4. Lack tool support

Easy meaning: No software tools to check if AI is ethical

Like building without a measuring tape:

- Builder: Is this wall straight?
- Current: I think so, looks okay
- Needed: Let me check with a level

For AI:

- Developer: Is our hiring AI fair?

- Currently: I think so, we tried
- Needed: Let me run the fairness checker

5. Lack lifecycle traceability

Lifecycle = From idea to thrown-away

- **Easy example:** Like tracking a car from design → factory → road → junkyard

Traceability = Following the breadcrumbs

- **Easy example:** If food makes you sick, you trace it back to the farm
- **For AI:** If AI is unfair, can we trace WHY?
 - Bad data? Bad programming? Bad testing?
 - Currently: Hard to trace, like finding a needle in a haystack

9- Dataset information must be properly explained and linked with the justification of the models used, since the system relies on both datasets and models for its operation.

Ethical Requirement	Best Dataset-Model Pair	Why This Pair Works
Fairness	COMPAS + Random Forest	RF's feature importance reveals racial bias patterns
Transparency	German Credit + Decision Tree	Tree rules show exactly why loans are approved/denied
Explainability	UCI Adult + Logistic Regression	Coefficients translate to "age increases income chance by X%"
Accountability	Lending Club + All Models	Multiple models provide cross-validation for audit trails

Use Case	Dataset	Model	Ethical Focus
Loan Fairness Audit	German Credit	Logistic Regression	Fairness (coefficient bias)
Hiring Transparency	UCI Adult Income	Decision Tree	Transparency (clear rules)
Medical Bias Detection	COMPAS	Random Forest	Fairness & Explainability

Marketing Accountability	Bank Marketing	All three	Comparative ethics analysis
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GEC Members

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2	Dr. Maqbool Khan	Co-Supervisor	School of Computing Sciences
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